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LEAD: Literature Enhanced Ab Initio Discovery of Nitride Dusting Layers for Enhanced Tunnel Magnetoresistance and Lower Resistance Magnetic Tunnel Junctions

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Magnetic tunnel junctions (MTJs) using magnesium oxide (MgO) tunnel barriers face challenges of high resistance-area product (RA) and low tunnel magnetoresistance (TMR). To discover alternative materials, Literature Enhanced Ab initio Discovery (LEAD) is developed as a framework that combines language models with first-principles screening. A domain-specific Word2Vec model is used to extract correlations from materials science literature, identifying promising candidates such as hybrid nitride-based barriers. Additionally, a Bidirectional Encoder Representations from Transformers based masked language model is trained on the same corpus to capture deeper contextual relationships, identifying tantalum nitride (TaN), vanadium nitride (VN), and titanium nitride (TiN) as candidate barrier materials. These predictions are subsequently evaluated using density functional theory (DFT) simulations in QuantumATK, benchmarking the predicted materials' TMR and RA against MgO and scandium nitride (ScN). Results show that MTJs featuring a monolayer dusting of ScN, or monolayer or bilayer dusting of TiN, on either side of MgO have similar or lower RA while achieving higher TMR than pure MgO junctions. LEAD offers a scalable method for discovering tunnel barriers and highlights nitride dusting layers as promising for next-generation MTJ performance.

1. Introduction

Magnetic tunnel junctions (MTJs) are critical components in numerous applications, including hard disk drive read heads,[1–3] nonvolatile memory,[4] radiofrequency (RF) oscillators,[5,6] magnetic random-access memory (MRAM),[7] in-memory computing,[8–10] and emerging computational architectures such as neuromorphic computing.[11,12] MTJs consist of two ferromagnetic layers separated by a thin insulating tunnel barrier, typically magnesium oxide (MgO), exhibiting high or low resistance based on spin-dependent tunneling in antiparallel or parallel magnetization configurations, respectively. The degree of spin-polarized tunneling through the tunnel barrier directly impacts key device figures of merit, such as tunnel magnetoresistance (TMR) and resistance-area product (RA).

Spin-dependent tunneling behavior was first exhibited in MTJs with an amorphous aluminum oxide (Al_2O_3)[13] barrier at room temperature, demonstrating TMR <70%. As more materials were investigated, it was found that MgO[14–16] barriers outperform Al_2O_3 barriers, demonstrating a room temperature TMR >600%.[17] The crystalline heterostructure in Fe(001)/MgO(001)/Fe(001) MTJs allows for electron states with Δ_1 spin symmetry to be transmitted efficiently through the tunnel barrier. This results in spin-polarized tunneling, as states with other symmetries decay rapidly. While MgO remains the dominant tunnel barrier material due to its ability to enable coherent spin filtering through symmetry selection, it suffers from practical limitations when deployed at nanometer scales. MgO's large bandgap of 7.8 eV[18] requires the use of ultra-thin layers to maintain acceptable RA. This, in turn, introduces thickness variations and pinhole formation across a silicon wafer, creating inconsistencies in TMR and RA. Thicker epitaxial barriers become a challenge due to the lattice mismatch between Fe and MgO (3.7%),[19] where Fe becomes more stressed as MgO thickness increases.

The scalability and performance of the technologies using MTJs largely depend on achieving a higher TMR at a lower RA product;[1] thus, there is a growing interest in finding an

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alternative barrier material. Prime candidates include materials that offer a reduced bandgap while still being insulators, which are compatible with standard Fe alloy electrode interfaces. Several options have been explored, including ultrathin Mg and CoFe insertion layers,[20] single-crystalline LiF,[21] alkali halides such as NaCl,[22] Heusler-based junctions with Co_2FeAl electrodes and a cation-disordered spinel Mg–Al–O barrier,[23] and 2D van der Waals materials such as MoS_2[24] and hBN.[25] Other directions include composite oxides such as CoO–ZnO,[26] high-entropy oxides like LiTiMgAlGaO,[27] small bandgap oxides such as $MgGa_2O_4$,[28] and rock-salt ScN.[29] However, achieving a higher TMR while maintaining a low RA remains a significant challenge and is the focus of ongoing research effort.

Conventional methods for exploring the vast materials space are often costly and rely on intuition-driven candidate selection, followed by time intensive first-principles screening and materials growth optimization, resulting in slow and limited progress. This presents the opportunity to leverage emerging machine learning and artificial intelligence techniques to guide the materials search.

The application of natural language processing (NLP) and large language models (LLMs) in materials science has expanded rapidly in recent years. Recent work emphasizes not only their broad capabilities but also the challenges associated with computational expense, hardware requirements, and ensuring reliability in research workflows.[30–35] LLMs have shown promising results in spite of these limitations. Their applications span property prediction,[36] molecule discovery for molecule-caption translation,[37] prediction of accurate atomization energies,[38] high-resolution structure protein prediction,[39] and autonomous chemical research.[40]

In parallel, distributed representations such as Word2Vec,[41] GloVe,[42] and subword embedding models[43] offer lightweight yet powerful alternatives for domain-specific applications. These methods have enabled discovery of new materials,[44] prediction of solar-chargeable batteries,[45] identification of harmful substances in food,[46] and even therapeutic assessments.[47] Transformer-based text-mining has also been applied to extract structured information from large corpora.[48]

Building on this momentum, domain-specific frameworks such as MatExpert,[49] LLMatDesign,[50] and goal-driven LLM agents[51] highlight how specialized pipelines can couple LLMs with simulation tools, curated datasets, and agent-based reasoning for hypothesis generation and autonomous materials discovery.

In the fields of magnetic materials and spintronics, due to the large collection of existing publications relevant to magnetism, these models can decipher connections within the research and can sift through the large amount of data which would be difficult for a human to process. This is especially relevant considering various materials being discussed in many different contexts across these publications. The usefulness of LLMs for spintronics material predictions has been proven through recent work by Sayed et al.[52] that predicted new and efficient spin–orbit torque materials with a high spin Hall angle that were then experimentally verified. However, this previous work does not leverage advanced language models like Bidirectional Encoder Representations from Transformers (BERT)[53] or LLMs, which offer superior contextual understanding. Also, experimentally testing tunnel barrier materials is a long process, since growth and annealing conditions for crystalline matching to electrodes and electrode optimization for TMR measurement are also required. For tunnel barriers, verification via density functional theory (DFT) is necessary for rapid materials discovery before proceeding to experiments.

Here, we design Literature Enhanced Ab initio Discovery (LEAD), a scalable pipeline that integrates NLP techniques with DFT to uncover novel tunnel barrier materials by finding hidden patterns present in the scientific literature, followed by first principles validation. LEAD incorporates not only the semantic relationships extracted via Word2Vec,[41] but also leverages the transformer-based model MaterialsBERT[54] and the open-weight generative LLM LLaMa 3.[55] The down-selected list of predicted materials includes TaN, VN, TiN, and hybrid MgO-nitride tunnel barriers. We subsequently validate these predictions with DFT simulations, assuming Fe electrodes using QuantumATK[56] to calculate the TMR and RA benchmarked against MgO and ScN. MgO and ScN are used as benchmarks as these are previously studied with pertinent data. Of all the materials validated, the outstanding candidates are thin TiN and ScN dusting layers added to either side of MgO. While maintaining a consistent number of total layers, the TiN and ScN dusting structure TMR ratios exceeded that of the MgO benchmark structure by over 5,000% and 7,000%, respectively, while maintaining RA values similar to MgO. Additionally, when increasing the oxide layer in between the interface dustings, the TMR ratios increase. For eight total atomic layers, the Fe/1ScN/6MgO/1ScN/Fe structure has a greater TMR than the Fe/8MgO/Fe and Fe/1TiN/6MgO/1TiN/Fe structures which both share a similar TMR. However, at 10 layers, both dusting structures share a TMR roughly 7,000% greater than that of Fe/10MgO/Fe. At 12 layers, the Fe/1TiN/10MgO/1TiN/Fe structure has a TMR (32,310%) that is greater than both Fe/1ScN/10MgO/1ScN/Fe (29,270%) and Fe/12MgO/Fe (23,540%). For the TiN-based structures, we find that doubling the interfacial nitride layers while maintaining the same MgO thickness (Fe/2TiN/4MgO/2TiN/Fe) yielded a high TMR of 10,600%. By coupling LMs with DFT simulations, the LEAD framework demonstrates a new paradigm for data-driven materials discovery. It identifies viable alternatives to pristine MgO barriers to advance the search for next-generation spintronic materials.

2. Model Setup

An overview of the LEAD framework methodology is shown in **Figure 1**. In Figure 1a, the data is first collected from journals sourced from Institute of Electronics and Electrical Engineers (IEEE), American Physical Society (APS), Elsevier, and Nature. The abstracts are subsequently pre-processed to produce a materials science abstract corpus. Figure 1b depicts how we then used this materials science abstract dataset to train LMs. We employed Word2Vec to extract semantic relationships between materials and MTJ-relevant properties. Once its predictive capacity is saturated, we transition to a BERT model, trained using masked language modeling (MLM). We compare multiple transformer based models: BERT-base, SciBERT,[57] and MaterialsBERT, and find that MaterialsBERT achieves the best performance for this task. To reduce invalid predictions from MaterialsBERT, LLaMa 3

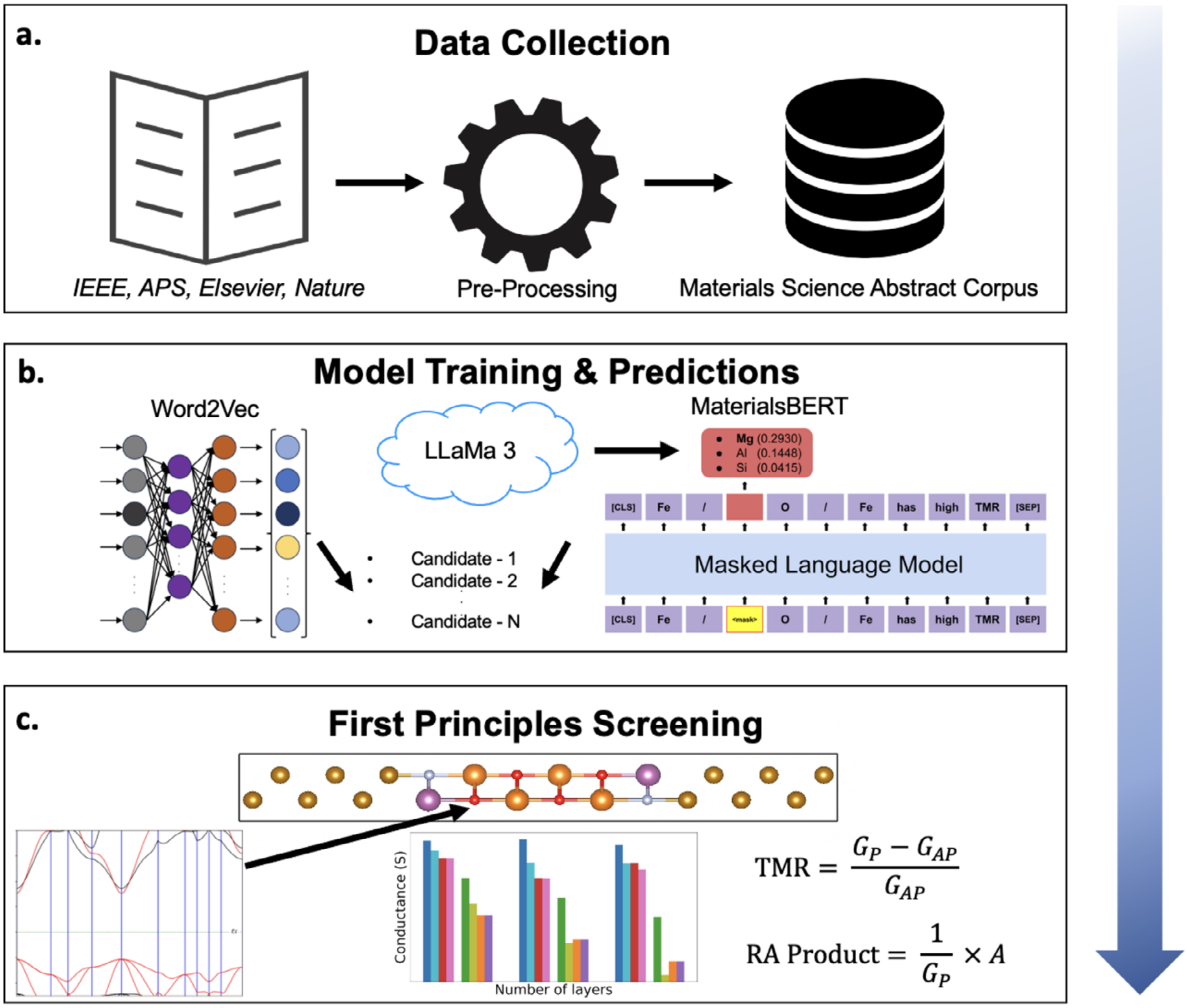

Figure 1. Overview of the LEAD workflow. a) Scientific abstracts are collected from peer-reviewed materials science, physics, and engineering journals (IEEE, APS, Elsevier, and Nature) and undergo preprocessing. b) Cleaned materials science abstract corpus is then used to train LLMs, which identify potential candidate materials. c) Candidates are subsequently evaluated using first-principles DFT calculations to assess their electronic and spin transport properties.

was first used to generate a broad list of chemically plausible tunnel barrier candidates. This list was subsequently filtered using the trained MaterialsBERT model, which ranks materials based on their relevance to MTJ applications. The output candidates were then evaluated via DFT calculations using QuantumATK, as shown in Figure 1c.

2.1. Training Corpus Preparation

The materials science abstract corpus for training consisted of over 1.1 million abstracts collected from peer-reviewed materials science, physics, and engineering journals published between 1965 and 2022. Sources included IEEE, APS, Elsevier, and Nature. These abstracts were merged into a unified corpus containing 220 million words and 1.5 billion characters prior to preprocessing. To minimize unintentional bias in the training text corpus, we avoided keyword-based searches during the collection of manuscript abstracts and refrained from structuring or classifying the data during preprocessing. Instead, we applied minimal preprocessing for standardization, including the removal of markup tags, punctuation, and extraneous symbols. The final cleaned corpus contained ≈140 million words and 880 million characters, which reflects a decrease of 36% in words and a 39% reduction in characters compared to the original dataset. The character count dropped due to the removal of non-informative markup, mathematical expressions, and special symbols, which inflate length without adding semantic value to the corpus. This preprocessing improved the corpus granularity and linguistic clarity, resulting in a measurable reduction in textual noise and complexity. Abstracts were chosen instead of full-text articles to prioritize information-dense content and reduce computational cost. The resulting materials science abstract for training provided a domain-specific language foundation which incorporated

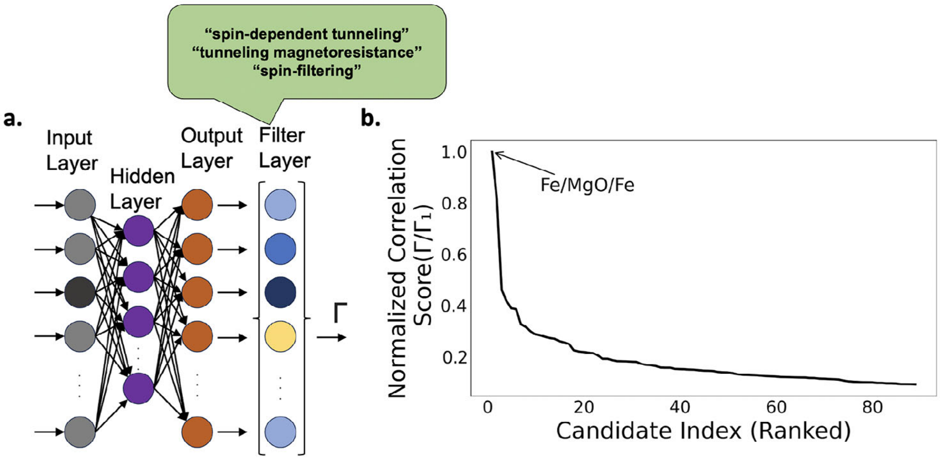

Figure 2. Architecture of the Word2Vec model and postprocessing workflow. a) The embedding model takes an input word represented as a sparse vector with a unique index set to one (grey input layer), passes it through a single hidden layer (purple), and outputs context predictions (orange) using negative sampling. The filter layer calculates the similarity score (Γ) between each material's learned embedding and the target query vector based on the domain-relevant phrases shown in the green speech bubble. b) Ranked list of candidate materials sorted by their similarity scores. All Γ values are scaled by maximum, Γ_1.

significant, peer-reviewed knowledge on materials, device structures, and spintronic phenomena relevant to MTJs.

2.2. Embedding Domain Knowledge for Semantic Screening

Initial candidate discovery leveraged an unsupervised word embedding approach inspired by the method described by Sayed et al.[52] In this approach, the Word2Vec embedding was implemented using the Gensim Python library (https:// radimrehurek.com/gensim/), utilizing the Skip-Gram[58] neural network architecture, visualized in **Figure** 2a. In Skip-Gram, a central target word is input to predict surrounding context words, capturing semantic relationships even for relatively rare materials or specialized terms. The key hyperparameters of the trained Word2Vec model were systematically optimized by assessing model stability, coherence of predicted semantic relationships, and known domain knowledge. The selected hyperparameters had a vector size = 300, window size = 10, negative sampling = 20, and minimum word count = 5.

The trained word embeddings allowed for the extraction of semantic associations between specific phenomena and materials relevant to MTJ applications. To identify candidate tunnel barrier materials, cosine similarity calculations of word embeddings were performed. Vectors corresponding to the domain-relevant phrases "spin-dependent tunneling," "tunneling magnetoresistance," and "spin-filtering" were used as semantic anchors. Cosine similarity between candidate material embeddings and these anchor vectors provided a quantitative metric (Γ) for ranking the relevance of the potential barrier materials. In its general form, Γ is defined over a set of embedding vectors $\vec{x}$ by

$$\Gamma = \sqrt{\prod_{i\neq j} \frac{\vec{x}_i}{\|\vec{x}_i\|} \cdot \frac{\vec{x}_j}{\|\vec{x}_j\|}} \tag{1}$$

which captures the joint alignment of all embeddings through their pairwise cosine similarities.

In this work, the set $\vec{x}$ consists of candidate material embedding $\vec{m}$ and three semantic anchors $\vec{a}_1$, $\vec{a}_2$, and $\vec{a}_3$. Restricting Equation (1) to material-anchor interactions yields the simplified form implemented in our model by

$$\Gamma = \cos(\vec{m}, \vec{a}_1) \cdot \cos(\vec{m}, \vec{a}_2) \cdot \cos(\vec{m}, \vec{a}_3) \tag{2}$$

Here, $\vec{m}$ denotes the embedding of the material, while $\vec{a}_1$, $\vec{a}_2$, and $\vec{a}_3$ denote the embeddings of the three semantic anchors of spintronic phenomena. Γ quantifies how strongly a material aligns with all three target concepts simultaneously. Candidates with high Γ values are more likely to exhibit tunneling behavior relevant to spintronic performance. To visualize the quality and distribution of model predictions, the normalized Γ scores of all ranked materials are plotted against their rank index in Figure 2b. The x-axis corresponds to the candidate rank and the y-axis shows Γ normalized to the top-ranked material. The resulting curve illustrates how quickly the semantic confidence drops beyond the top two candidates, Fe/MgO/Fe and CoFeB/MgO/CoFeB, respectively. A steep drop-off implies that only a limited subset of materials are strongly aligned with all three target phenomena. Candidate filtering also included domain-specific rules to exclude non-relevant terms, utilizing additional criteria based on chemical formula structure and known materials behavior. These filters ensured that retained candidates were chemically meaningful and semantically coherent within the context of spintronic applications.

2.3. Contextual Embedding with Transformer Models

While word embedding models like Word2Vec are effective for identifying new results through classification and clustering of static embeddings, they are less adept at capturing multiple contexts of the same materials and phenomena, particularly in large

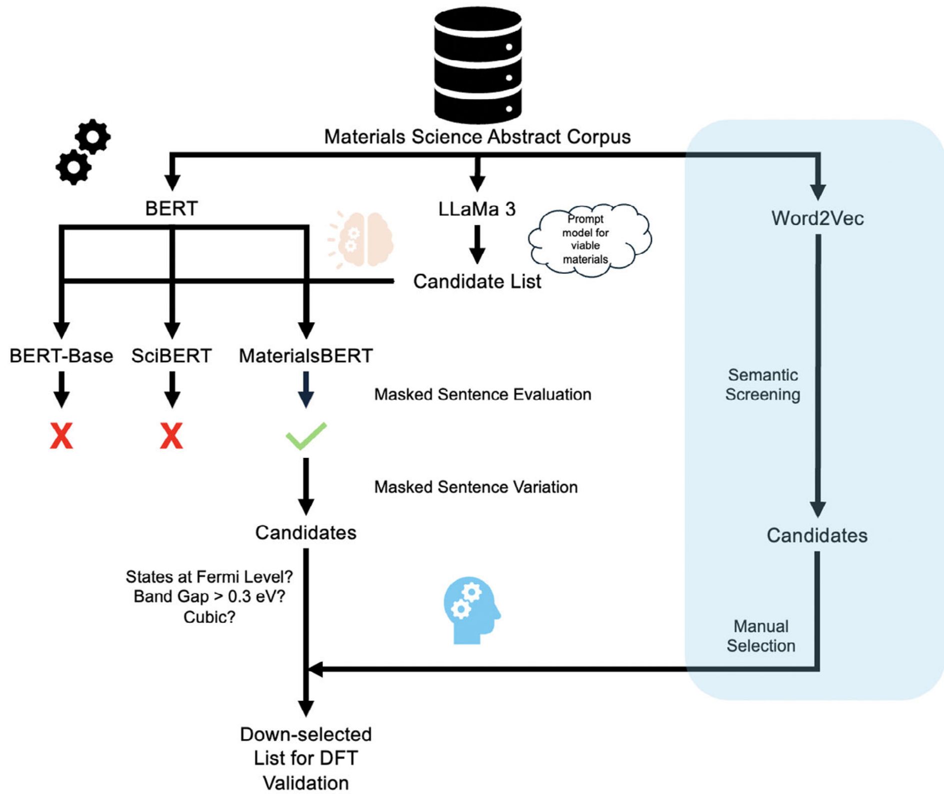

Figure 3. Pipeline utilized for filtering tunnel barrier material candidates using Word2Vec and MaterialsBERT masked language model coupled with prompt-based generative predictions. Blue shading shows the Word2Vec pipeline, in parallel with the coupled BERT and Llama 3 pipeline.

text corpora. Therefore, we adopt a model that generates contextual embeddings, enabling the capture of long-range dependencies and complex syntactic relationships critical for applications such as materials prediction. **Figure 3** situates the transformer-based approach within the overall LEAD pipeline. As shown, Word2Vec embeddings provide an initial semantic screen of materials, while a parallel branch leverages BERT-based models in conjunction with LLaMa 3 for refined candidate generation and filtering. For the transformer branch, contextual embeddings enable a more accurate representation of scientific abstracts.

To improve contextual accuracy beyond Word2Vec, we adopted a BERT-based transformer model, which uses bidirectional attention to incorporate information from both sides of a token. This is particularly useful for scientific abstracts, where term meaning often depends on surrounding clauses. We first used SciBERT, which was pretrained on scientific literature from Semantic Scholar, and further pretrained it on our domain specific corpus (the same used for Word2Vec) using the MLM objective. MLM randomly masks tokens and trains the model to predict them based on context, allowing it to learn contextualized embeddings for words and chemical entities. To enhance domain specificity, we then employed MaterialsBERT, pretrained from PubMedBERT,[59] on materials science texts from Elsevier, Wiley, Royal Society of Chemistry, American Chemical Society, Springer Nature, Taylor & Francis, and the American Institute of Physics. This model was also trained using MLM. The tokenizer vocabulary for MaterialsBERT was fixed from the original pretrained model checkpoint to reduce computational demand. For baseline comparison, we trained BERT-base, originally pretrained on BookCorpus and Wikipedia, on our corpus as well. To evaluate the models' domain understanding, we used a single masked-token prompt: "The most commonly used tunnel barrier in Fe-based MTJs due to its rock-salt structure and Δ_1 symmetry filtering is <mask>O." This prompt was tested across all models to assess their ability to predict suitable tunnel barrier materials. Final MLM training metrics, including loss, accuracy, and perplexity, are reported in **Table 1**, showing that all models converged successfully.

To train the transformer models, the AdamW optimizer with a learning rate of 2×10^{-5} and batch size of 16 was used. Training was performed for three epochs using the MLM objective with a maximum sequence length of 512 tokens. Default weight decay (0.01) and linear warmup were applied.

Table 1. Comparison of Evaluation Metrics Across Models.

Metric	SciBERT	MaterialsBERT	BERT-base
Evaluation Loss	1.20	1.20	1.21
Accuracy (%)	72.4	72.7	73.0
Perplexity	3.30	3.33	3.34
Training Loss	1.28	1.31	1.37

2.4. Generative Expansion and Candidate Refinement

Non-material mask predictions were common among results. In order to narrow the search space to just candidate barrier materials, Meta's generative model, LLaMa 3, was employed. As with the previous models, LLaMa 3 was fine-tuned on the same materials science abstracts corpus. Using prompt engineering, the trained LLaMa 3 model was queried to produce a comprehensive list of chemically plausible barrier materials based on clearly defined criteria. These included good band symmetry with Fe, high TMR, low RA, and effective symmetry filtering. This generative step provided an extensive list of candidate materials.

The generative output was then filtered using the trained MaterialsBERT model, where sentences containing candidate materials were masked. For each candidate barrier, we masked its chemical symbol inside an MTJ-specific sentence and computed the probability that the model would fill the blank with a given element token. Formally, for logits $\mathbf{z} \in \mathbb{R}^{|V|}$ at the masked position, the probability assigned to a token w_i is

$$P(w_i \mid \text{context}) = \frac{\exp(z_i)}{\sum_{j=1}^{|V|} \exp(z_j)} \tag{3}$$

Candidate elements were ranked by P, and only those that scored consistently high across multiple prompts were retained. Rather than generating arbitrary completions, the model evaluates the compatibility of each element with MTJ-relevant contexts based on its learned scientific understanding. Only materials that consistently rank within the top ten across multiple prompts are considered further. The final shortlisted materials were manually selected based on their scores and additional criteria: a bandgap >0.3 eV, no states at the Fermi energy E_f, and a cubic lattice system. The use of this filtering scheme reduced the chance of improbable candidates that did not fit within the sentence mask.

Each model in the LEAD framework contributes distinct strengths. Word2Vec captures broad corpus-level relationships that reveal chemically associated materials, while MaterialsBERT provides context-aware ranking of physically meaningful completions. LLaMA 3 introduces a wider generative reasoning capability that enables it to move beyond patterns seen in the training literature and propose chemically consistent material classes.

2.5. First-Principles Screening Setup

The supercell structures used in this study were constructed using QuantumATK. Periodic boundary conditions were employed in the lateral (x and y) directions and a finite boundary condition was deployed along the transport direction (z). The supercell consisted of a layered structure comprising Fe electrodes sandwiching various candidate tunnel barrier materials. To match with experimental Fe electrodes (lattice constant 2.866 Å[60]), the barrier layers were compressed appropriately, inducing a strain that varied depending on the lattice mismatch. The supercells were built using fractional coordinates derived from literature and relaxed with QuantumATK's Linear Combination of Atomic Orbitals calculator. During relaxation, a spin generalized gradient approximation (SGGA) using the Perdew–Burke–Ernzerhof (PBE) exchange-correlation functional was employed. Structural optimizations were conducted using the Limited-memory Broyden Fletcher Goldfarb Shanno algorithm, converging atomic forces to below 0.05 eV $\forall^{-1}$. To ensure realistic interface conditions, rigid constraints were applied to the outermost electrode layers, while the electrode interface and tunnel barrier layers were allowed to fully relax.

The electronic bandstructures of the barrier materials and electrode-barrier interfaces were calculated using QuantumATK. Monkhorst-Pack k-point grids were set sufficiently dense (13× 13 × 13 for bulk calculations) to ensure accurate representation of electronic states near E_f. Bandstructures were computed along high-symmetry paths in reciprocal space with QuantumATK's built-in bandstructure module. This captures bands both below and above E_f. Standard DFT calculations using the PBE exchange-correlation functional are known to significantly underestimate bandgaps in transition metal nitrides due to insufficient localization of partially filled d orbitals. To correct for this to more accurately reflect the experimental bandgap, Hubbard potential (U) corrections were applied to the d orbitals of the transition metal atoms in the candidate materials. ScN was used as a benchmark system due to the availability of experimental bandgap measurements, with the initial version of the inaccurately computed structure shown in **Figure 4a**. To achieve its indirect bandgap of 0.9 eV, a systematic sweep of U correction values was performed, ranging from 3 to 10.5 eV. Lower U values led to incomplete orbital localization. By contrast, a U correction value of 7 eV successfully opened an indirect bandgap close to 0.9 eV as shown in Figure 4b. The projection scheme in QuantumATK, which uses explicit atomic projections rather than pseudopotential-based methods, typically requires slightly higher U values over this range.

Complex bandstructures of the tunnel barrier configurations were calculated in QuantumATK to quantify state decay at Γ and X points. The intersection of the complex band energies with E_f determines the barrier material's rate of decay. Calculations were performed on heterostructure geometries with periodic boundary conditions. Complex bandstructure spectra were computed over an energy window from −4 to +6 eV relative to the self-consistent Fermi level, sampled at 3001 points. The imaginary part of the out-of-plane wavevector, Im(k), was extracted at each energy to determine decay constants k.

Spin-dependent electron transport properties were calculated using QuantumATK's non-equilibrium Green's function (NEGF) formalism combined with DFT. Device configurations were constructed from the previously relaxed supercells by attaching semi-infinite Fe electrodes to the left and right of the central tunnel barrier region. Separate transport calculations were carried out for the parallel (P) and antiparallel (AP) magnetic configurations of the Fe electrodes.

The spin-polarized exchange-correlation potential was treated using SGGA in the PBE form. To capture spin asymmetry and enforce magnetic ordering, initial spin states were explicitly set on the Fe atoms in both electrodes. For the parallel configuration, spins were aligned ferromagnetically across the junction, while in the antiparallel case, spins in the right electrode were inverted to model an opposing magnetization state. Nonmagnetic atoms within the barrier were initialized with zero spin.

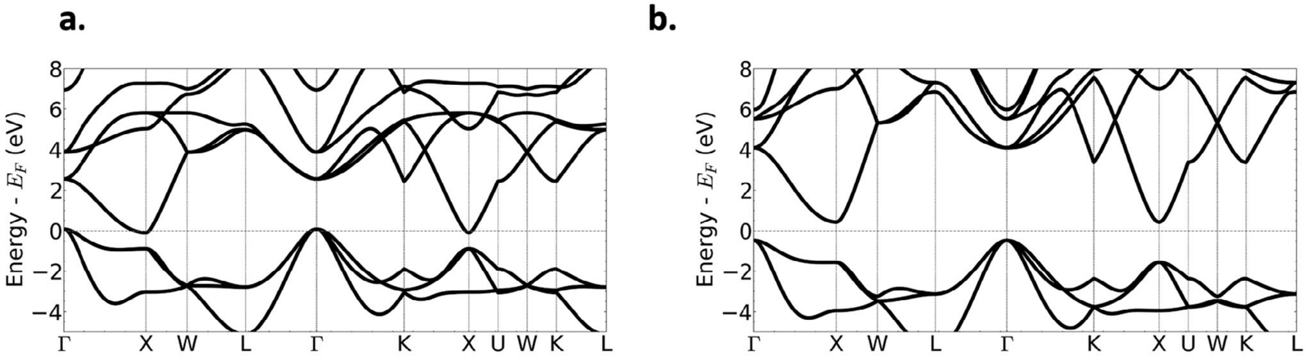

Figure 4. ScN band structures. a) Band structure of ScN calculated with the PBE functional, showing an underestimated indirect gap. b) Band structure of ScN after applying a 7 eV Hubbard potential to the Sc 3d orbitals, yielding the experimentally-observed 0.9 eV gap.

Transport calculations were performed using a dense Monkhorst-Pack grid (151 × 151) in the transverse plane to ensure an accurate Brillouin zone integration. The transmission spectra at zero bias were computed using QuantumATK's transmission module. Total conductance for each configuration was obtained by summing the spin-resolved transmission at E_f for spin-up and spin-down channels. The TMR was computed from the conductances in parallel (G_P) and antiparallel (G_{AP}) configurations using $\mathrm{TMR} = \left(\frac{G_P - G_{AP}}{G_{AP}}\right) \times 100\%$. The RA is defined as $RA = \frac{1}{G_P} \cdot A$, where A is the in-plane cross-sectional area of the simulation supercell. These computational procedures allowed for a detailed theoretical understanding of material-specific tunneling properties.

A summary of the DFT setup specifications can be found in Table S1 (Supporting Information).

3. Results

3.1. Model Predictions

The Word2Vec model highlighted several intriguing predictions. Thirty-eight material predictions were chosen based on their Γ values (cutoff Γ = 0.0421), see Table S2 (Supporting Information). The model strongly favored MgO-based configurations, which provides validation of the method. The highest-ranking candidates were "Fe/MgO/Fe" and "CoFeB/MgO/CoFeB," with semantic similarity Γ scores of 0.2769 and 0.2257, respectively. These results reflect the model's capability to recognize well-established MTJ structures associated with the domain-relevant phrases ("spin-dependent tunneling," "tunneling magnetoresistance," and "spin-filtering"). A significant drop in similarity scores was observed for subsequent materials, further underscoring the model's effectiveness in distinguishing known MTJ configurations. Among the predicted materials, barriers featuring niobium were frequently identified, most notably the hybrid barrier "NbN/MgO/NbN." This prevalence is plausibly driven by the extensive superconducting tunnel junction literature on Nb and NbN-based Josephson junctions, where MgO as well as other oxide barriers are often discussed in the same semantic context. After human-in-the-loop down selection, candidates of interest from the trained Word2Vec model were hybrid "NbN/MgO/NbN" and "DyN."

Table 2. Top ten predictions for masked sentence: "The most commonly used tunnel barrier in Fe-based MTJs due to its rock-salt structure and Δ_1 symmetry filtering is <mask>O." Predictions for cation elements (i.e., the masked position before "O") are shown for the three pretrained BERT models, along with their respective probabilities (P), and compared against MaterialsBERT with no further pretraining on our corpus as a control experiment (rightmost column).

Rank	BERT-base (P)	SciBERT (P)	MaterialsBERT (P)	MaterialsBERT (control) (P)
1	V (0.059)	Fe (0.347)	Mg (0.298)	Fe (0.042)
2	B (0.038)	C (0.071)	V (0.171)	Al (0.035)
3	C (0.026)	Al (0.053)	Al (0.089)	W (0.016)
4	K (0.009)	Si (0.041)	Ta (0.058)	Mg (0.016)
5	Li (0.004)	Cu (0.040)	W (0.042)	Cu (0.013)
6	Si (0.003)	V (0.037)	C (0.041)	V (0.009)
7	Al (0.002)	Ti (0.034)	Cu (0.021)	In (0.008)
8	Ti (0.002)	Hf (0.030)	Ti (0.015)	Si (0.005)
9	W (0.002)	Be (0.019)	Si (0.014)	Ga (0.004)
10	In (0.002)	In (0.009)	Fe (0.010)	C (0.004)

To evaluate the effectiveness of transformer models specifically trained on MTJ-relevant literature, a two-step approach was adapted. First, the trained LLaMa 3 model was prompted to produce a comprehensive list of chemically plausible cation candidates, spanning alkaline earth metals, p-block elements, metalloids, transition metals, and lanthanides (see Table S3, Supporting Information). This list was used strictly as a filtering scheme to constrain the masked predictions and avoid invalid chemical outputs. With this candidate set fixed, we then compared the performance of BERT-base, SciBERT, and MaterialsBERT on the same masked sentence. The ranked predictions are summarized in **Table 2**. Among the tested models, the pretrained MaterialsBERT most accurately identified "Mg" with the highest probability P, whereas the control experiment without pretraining on the materials science abstract corpus produced only a very low P, highlighting the importance of domain adaptation.

The trained MaterialsBERT suggested promising materials. We constructed masked sentences based on domain knowledge

Table 3. Down-selected materials for MTJ tunnel barriers and their source of prediction.

Material	Source
MgO	Benchmark
ScN	Benchmark
NbN/MgO/NbN	Word2Vec
DyN	Word2Vec
TiN	LLaMa 3 + MaterialsBERT
TaN	LLaMa 3 + MaterialsBERT
VN	LLaMa 3 + MaterialsBERT
BaO	LLaMa 3 + MaterialsBERT
ZnO	LLaMa 3 + MaterialsBERT

of MTJ structures, symmetry filtering, and material requirements. Prompts included phrases such as "Ferromagnetic tunneling devices using Fe/<mask>N/CoFeB stacks are expected to sustain large magnetoresistance at room temperature while offering high thermal stability" and "Among oxides, <mask>O barriers lattice-match well with Fe(001), enabling efficient spin polarized tunneling." For full list, see Tables S4 and S5 (Supporting Information). These were designed to elicit chemically relevant elements based on the masked token's contextual surroundings. Materials that were consistent among the top 10 for *P* value and passed the manual screening were chosen. Elements Cu, Ta, V, Ti and Ba were selected because they appeared consistently across multiple prompts in appropriate chemical contexts (e.g., <mask>O, <mask>2O, or <mask>N) and reflect realistic compounds such as Cu_2O, BaO, TiN, TaN, and VN. These materials were most common from the MaterialsBERT model when prompting for materials that had good spin symmetry and lattice matching with Fe. The materials were further vetted for structural plausibility and suitability as tunnel barriers based on known crystal structure and electronic structure. Although TiN has no bandgap, its lattice constant (4.25 Å)[61] is similar to that of MgO (4.21 Å),[62] prompting its inclusion among the further vetted materials. The down-selected set of materials across all models is shown in **Table 3**. DFT calculations were subsequently performed to validate whether these predictions could become suitable improvements for MTJs.

Comparative evaluation of the three NLP models further clarifies their complementary roles. The ablation experiments of prompting each model on their own reveal their different strengths. Compared with MaterialsBERT, Word2Vec primarily surfaced co-mentioned oxides without distinguishing crystal symmetry. LLaMA 3 broadens exploratory space through contextual generalization, as demonstrated by its completion of the same masked prompt used for evaluating the BERT models (see Table S6, Supporting Information). LLaMA 3 correctly predicts MgO and ranks chemically consistent alternatives such as AlO_x, TiO_x, and HfO_x, showing that it can move beyond corpus-level statistical patterns and generate physically meaningful predictions. This underscores the role of LLaMa 3 as an exploratory model in the pipeline whose generative output augments the context-aware ranking provided by MaterialsBERT. Their combined use thus enables both creativity and physical validity in literature-driven materials discovery.

3.2. First-Principles Analysis

The bandstructures of the two benchmarks and two candidate materials are shown in **Figure 5**. MgO is presented first as a benchmark in Figure 5a, followed by ScN (Figure 5b), NbN (Figure 5c), and TiN (Figure 5d). Figure 5b,c reveals that the bandstructure of NbN resembles that of ScN despite its metallic character due to their shared rocksalt structure and similar orbital symmetries. The additional d-electrons in Nb shifts both the conduction and valence bands downward in energy. As a result, NbN retains similar band dispersions as ScN, with the conduction band minimum at X and a W–L–Γ ridge, but with its Fermi level pinned within the conduction band. TiN (Figure 5d) exhibits a bandstructure similar to NbN, suggesting that it can also supports symmetry-filtered Δ_1 and Δ_2' tunneling channels. Combined with its improved interface stability, TiN emerges as a strong candidate for hybrid MgO-based tunnel barriers. Bandstructures for DyN, TaN, VN, BaO, and ZnO are found in Figure S1 (Supporting Information). Hubbard U corrections were applied only where necessary to correct for self-interaction errors in localized d-states. Among the down-selected materials, ScN exhibits partially filled Sc 3d orbitals and a small bandgap that is underestimated in GGA. As outlined in the methods, a U value of 7 eV was therefore applied to reproduce the experimental indirect $E_g \approx 0.9$ eV. All other candidate materials (MgO, TiN, NbN, TaN, VN, BaO, ZnO) are either metallic or wide-gap ionic compounds for which electron correlation effects are negligible.

Calculated conductances and TMR ratios for the significant predicted candidates across Word2Vec and MaterialsBERT models are summarized in **Table 4**. The six-layer MgO MTJ is provided as a benchmark for the other structures, with a TMR ratio of 7, 180%. As expected, DFT predicts much higher TMR than experimental values, and similarly high values are needed for the new materials to be technologically relevant.[15] The secondary benchmark, six-layer ScN, shows a more modest TMR of 530%, while pristine six-layer TiN performs comparably at 580%. In contrast, pristine NbN is significantly less effective, with a TMR of only 30%, and BaO reaches 400%, both well below MgO. Among the pristine candidates, six-layer ZnO is notable for achieving 6,600%, only 500% below MgO, making it one of the strongest alternatives. At the other extreme, VN and TaN exhibit low TMR ratios, 100% and 80%, respectively. DyN also falls short, with 140% for the pristine barrier and 220% for the hybrid structure.

The hybrid nitride MgO-based structures highlight how dusting layers influence performance. The Fe/1NbN/4MgO/1NbN/Fe configuration fell short of the MgO benchmark due to factors such as a high density of states at E_f, Fermi-level pinning, and reduced tunability from semimetallic interfaces. In contrast, replacing NbN with ScN or TiN eliminated many of these challenges. Remarkably, both Fe/1ScN/4MgO/1ScN/Fe and Fe/1TiN/4MgO/1TiN/Fe exceed the pristine MgO benchmark, achieving TMR values of 14,300% and 12,000%, respectively. The success of ScN and TiN, combined with their bandstructure similarity to NbN, motivated the extension of this strategy to other nitride predictions. Consistent with this rationale, additional nitride hybrids such as Fe/1TaN/4MgO/1TaN/Fe and Fe/1VN/4MgO/1VN/Fe also outperformed the NbN hybrid, though their ratios remained below pristine MgO. These results

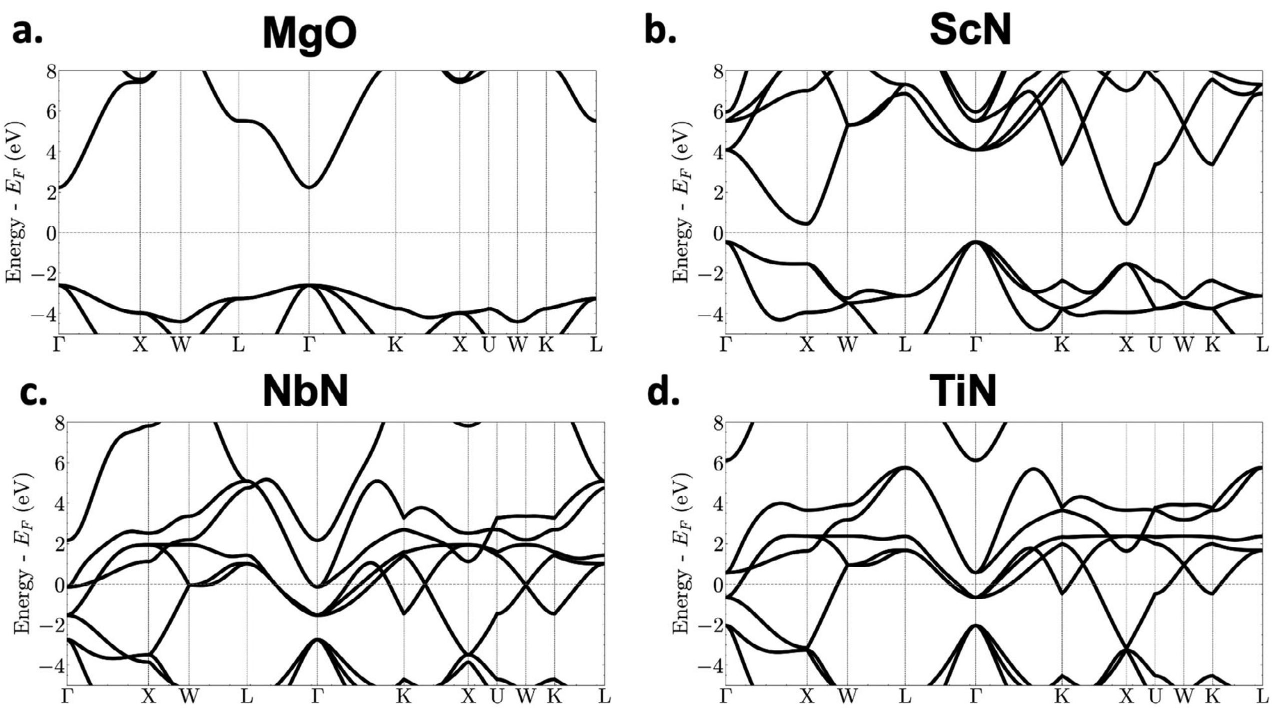

Figure 5. Bulk band structures of two down-selected tunnel barrier candidates vs. benchmarks. a) MgO is a wide-gap insulator used as the MTJ baseline. b) ScN exhibits an indirect Γ to X gap of 0.9 eV after +U of 7 eV. c) Semimetal NbN with bands crossing E_f, giving high DOS. d) TiN is a metallic nitride material that offers a good lattice match to Fe/MgO.

underscore the dominance of nitride materials as effective dusting layers for enhancing TMR.

Motivated by these observations of the hybrid barriers, different hybrid barrier configurations incorporating ScN and MgO layers were investigated (**Figure 6**). Among the configurations tested, the monolayer ScN dusting (Figure 6a) performed the best. With a thicker layer of ScN at the interface while maintaining the total number of layers as 6 as shown in Figure 6b, the TMR ratio of the bilayer-dusting MTJ resulted in 820%. When increasing the thickness of the MgO core (Figure 6c) the TMR

Table 4. Spin-resolved conductances for the parallel (P) and antiparallel (AP) states, and the resulting TMR.

Layers	$G_P^{\uparrow}$ (S)	$G_P^{\downarrow}$ (S)	G_P (S)	$G_{AP}^{\uparrow}$ (S)	$G_{AP}^{\downarrow}$ (S)	G_{AP} (S)	TMR (%)
Pristine barriers							
Fe/MgO(6)/Fe	1.18e-08	3.91e-10	1.22e-08	8.36e-11	8.34e-11	1.67e-10	7,180
Fe/ZnO(6)/Fe	6.25e-07	9.00e-08	7.15e-07	6.57e-09	4.10e-09	1.07e-08	6,600
Fe/TiN(6)/Fe	1.06e-05	1.29e-06	1.18e-05	8.79e-07	8.78e-07	1.76e-06	570
Fe/ScN(6)/Fe	1.65e-06	4.56e-07	2.10e-06	1.66e-07	1.66e-07	3.31e-07	530
Fe/BaO(6)/Fe	2.82e-07	2.28e-08	3.05e-07	3.01e-08	3.04e-08	6.05e-08	400
Fe/DyN(6)/Fe	1.32e-05	3.81e-06	1.70e-05	3.59e-06	3.62e-06	7.21e-06	140
Fe/VN(6)/Fe	1.06e-05	4.41e-06	1.50e-05	3.83e-06	3.83e-06	7.66e-06	100
Fe/TaN(6)/Fe	9.61e-06	4.83e-06	1.44e-05	4.11e-06	4.10e-06	8.21e-06	80
Fe/NbN(6)/Fe	1.57e-05	6.80e-06	2.25e-05	8.80e-06	8.81e-06	1.76e-05	30
Hybrid nitride structures							
Fe/ScN(1)/MgO(4)/ScN(1)/Fe	8.12e-09	4.34e-09	1.25e-08	5.11e-11	3.57e-11	8.68e-11	14,260
Fe/TiN(1)/MgO(4)/TiN(1)/Fe	8.73e-09	2.41e-11	8.75e-09	3.62e-11	3.62e-11	7.24e-11	12,000
Fe/VN(1)/MgO(4)/VN(1)/Fe	1.01e-07	1.90e-10	1.01e-07	1.20e-09	1.17e-09	2.37e-09	4,170
Fe/TaN(1)/MgO(4)/TaN(1)/Fe	1.47e-08	7.93e-11	1.48e-08	1.90e-10	1.90e-10	3.80e-10	3,790
Fe/NbN(1)/MgO(4)/NbN(1)/Fe	1.60e-08	1.51e-10	1.62e-08	5.17e-10	5.17e-10	1.03e-09	1,470
Fe/DyN(1)/MgO(4)/DyN(1)/Fe	1.55e-07	1.21e-08	1.67e-07	2.76e-08	2.51e-08	5.27e-08	220

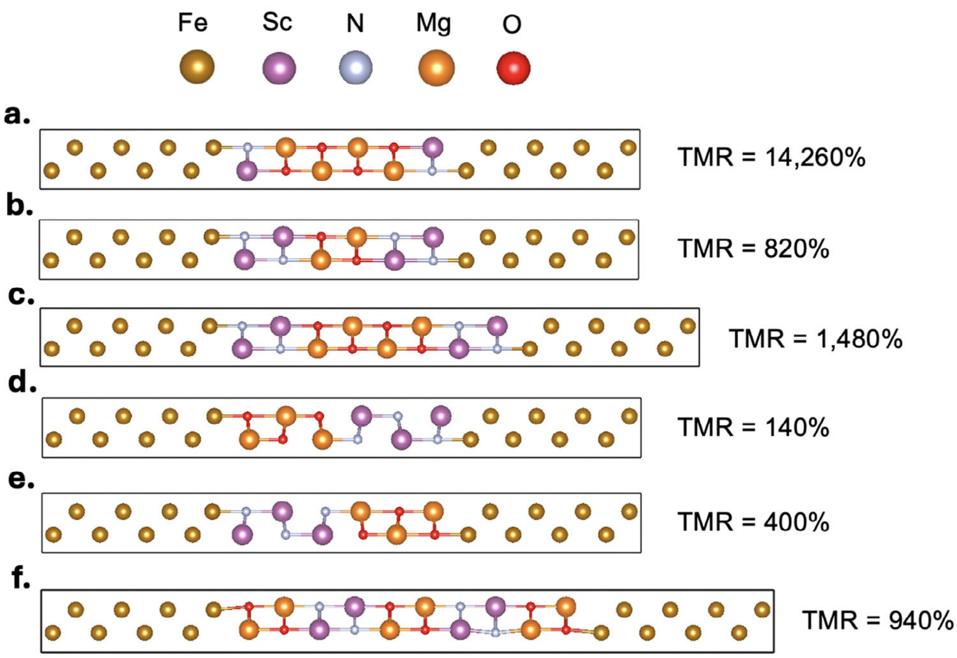

Figure 6. ScN-MgO hybrid tunnel barrier configurations (Fe electrodes). TMR ratio of each configuration shown on right. a) Monolayer-dusting design Fe/1ScN/4MgO/1ScN/Fe (six total barrier layers), which yields the highest simulated TMR among the hybrids. b) Bilayer-dusting design Fe/2ScN/2MgO/2ScN/Fe with the same total thickness, illustrating the trade-off between added nitride and spin-filter selectivity. c) Thickened-core design Fe/2ScN/4MgO/2ScN/Fe (eight layers) that restores a broader MgO section while retaining interfacial ScN. d) MgO-favored half and half Fe/3MgO/3ScN/Fe, grouping the oxides and nitrides into contiguous blocks. e) ScN-favored half and half Fe/3ScN/3MgO/Fe, blocking the nitride first and then the oxide. f) Alternating superlattice Fe/2MgO/2ScN/2MgO/2ScN/2MgO/Fe (twelve layers) that repeats two-layer MgO and ScN units. All models are oriented along the (001) direction in the cubic rock-salt structure, with periodic boundary conditions applied along the transport (z) direction. Fe electrodes (gold) extend semi-infinitely on both sides of the tunnel barrier, which they clamp along the transport axis. Within the barrier region, purple and orange spheres represent Sc and Mg atoms, gray and red denote N and O atoms, and electron transport proceeds along the z-axis (right direction).

ratio increased to 1,480%. Figure 6d depicts how a tunnel barrier of half MgO and half ScN was constructed, exhibiting a low TMR of 140%. In Figure 6e, however, when switching the order of ScN and MgO, the TMR ratio increases to 400%. When alternating the layers of ScN and MgO in the the superlattice as shown in Figure 6f, the TMR ratio resulted in 940%.

Encouraged by the enhanced TMR achieved with a single-layer ScN dusting, we substituted TiN into the same architectures (see Figure S2, Supporting Information), where the monolayer TiN dusting similarly performed the best. **Figure 7** compares the thickness dependence of three stacks: pure MgO, ScN-dusted MgO, and TiN-dusted MgO. Several trends emerge, one of these being that across all thicknesses the nitride-dusted hybrids outperform pristine MgO. At six total layers, the addition one TiN atomic layer to each interface (Fe/1TiN/4MgO/1TiN/Fe) boosts TMR from 7,180% (Fe/6MgO/Fe) to 12,000%, while ScN (Fe/1ScN/4MgO/1ScN/Fe) lifts it still higher to 14,230%. The improvement stems from the nitride's narrower effective barrier height, which raises majority-spin conductance, without compromising Δ_1 filtering from MgO. As the total number of barrier layers is thickened from six to eight layers, TMR in the Fe/1ScN/6MgO/1ScN/Fe hybrid continues to climb steeply, reflecting MgO's restored symmetry selectivity. Fe/1TiN/6MgO/1TiN/Fe levels off with a ratio similar to Fe/8MgO/Fe but less than the ScN hybrid. Once the barrier exceeds ten layers the situation reverses: Twelve layers Fe/1TiN/10MgO/1TiN/Fe achieves 32,310%, outperforming both Fe/1ScN/10MgO/1ScN/Fe (29,270%) and Fe/12MgO/Fe (23,540%).

For ultrathin junctions aimed at the lowest possible RA, ScN remains the superior dusting choice due to its higher initial TMR

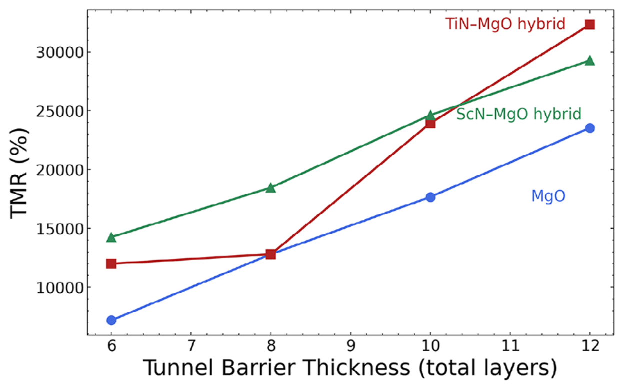

Figure 7. Comparison of a pure Fe/MgO/Fe junction (blue circles) against nitride-dusted structures in which a single ScN or TiN monolayer is placed at each interface: Fe/1ScN/*x*MgO/1ScN/Fe (green triangles) and Fe/1TiN/*x*MgO/1TiN/Fe (red squares). The MgO thickness *x* is systematically increased from 4 to 10 layers, giving overall barrier thicknesses between 6 and 12 layers.

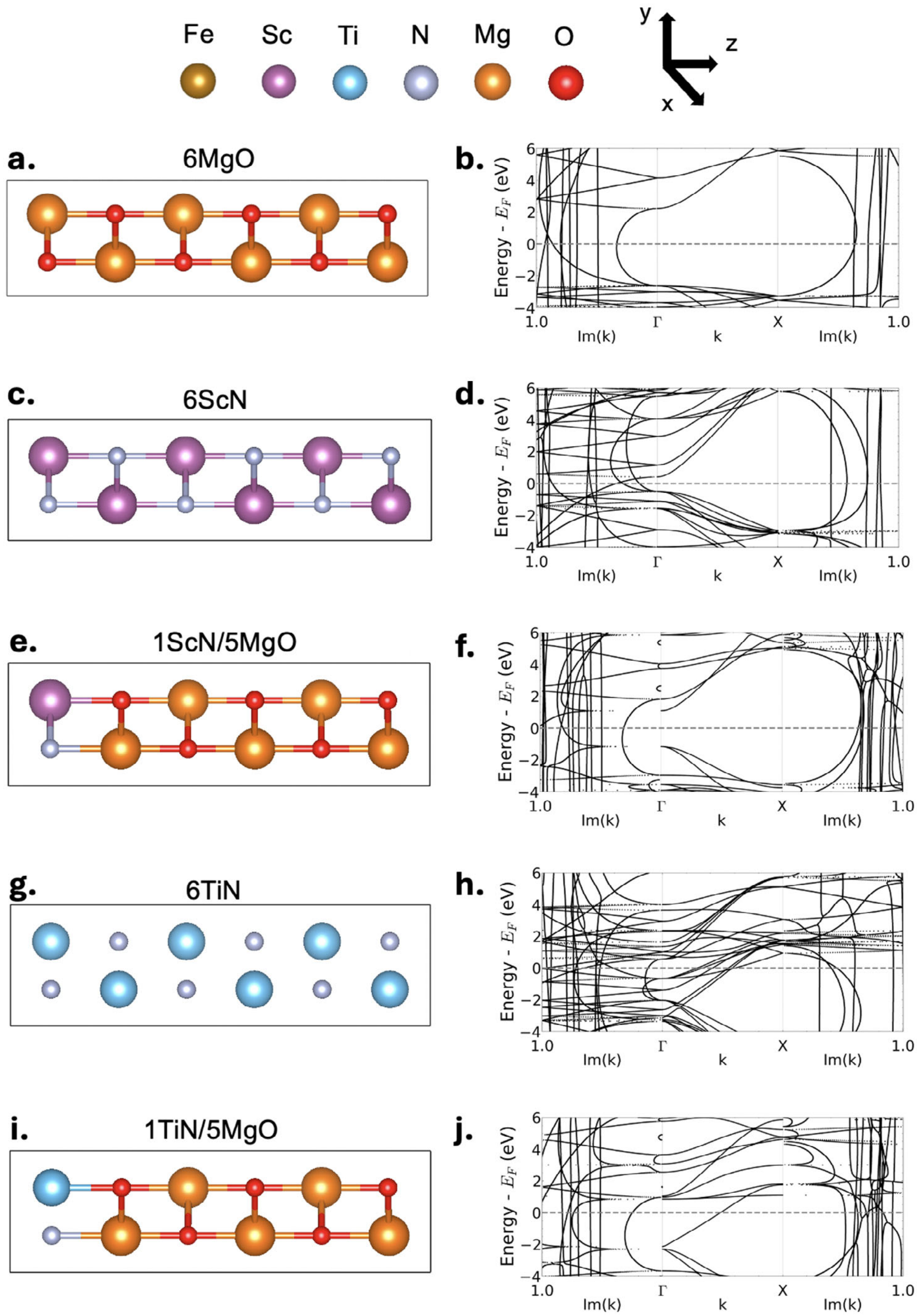

Figure 8. Tunnel barrier heterostructures with corresponding complex bandstructures. Complex band structures are obtained by extending from the Γ and X points into imaginary *k*-space, with symmetry-resolved real bands shown in the central region. a) Six-layer MgO heterostructure and b) its complex bands; c) six-layer ScN and d) its complex bands; e) 1ScN/5MgO and f) its complex bands; g) six-layer TiN and h) its complex bands; i) 1TiN/5MgO and j) its complex bands. All periodic structures are oriented along the (001) direction in the cubic rock-salt configuration. Purple, orange, and blue spheres represent Sc, Mg, and Ti atoms, respectively, while gray and red denote N and O atoms. The electron transport direction is along the z-axis (rightward).

and minimal minority-spin leakage. For moderately thick barriers, TiN becomes attractive because its TMR continues to rise even as the ScN plot begins to saturate. Moreover, TiN's electrical conductivity could help further suppress RA. This underscores how the optimal nitride dusting layer depends on the overall barrier thickness and target RA and TMR trade-off.

To better understand the enhanced performance of the ScN and TiN dusting layer MTJs, the complex bandstructures of relevant configurations are calculated in **Figure 8**. The tunneling transmission through an insulating barrier depends exponentially on the decay constant $|\kappa|$ (imaginary wavevector) extracted from the complex bands, such that smaller $|\kappa|$ values correspond

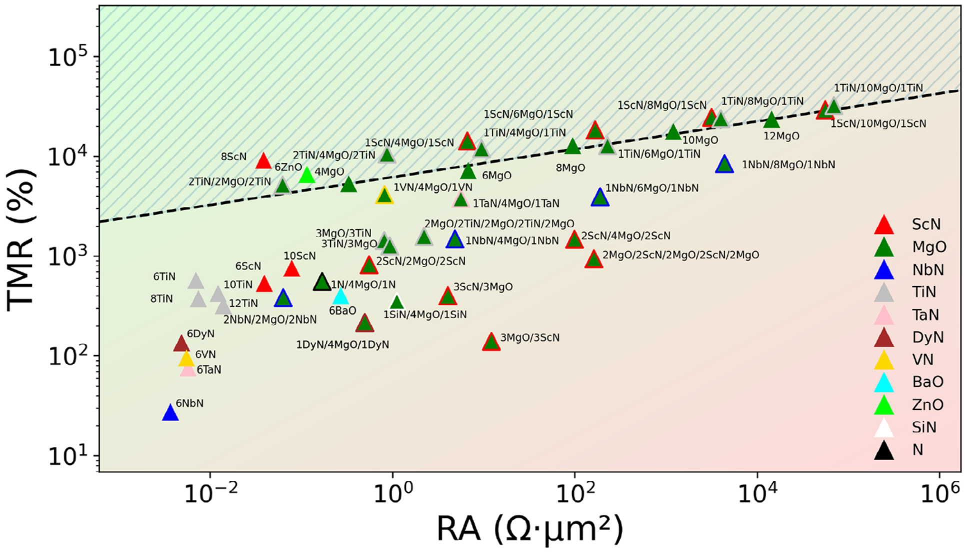

Figure 9. TMR vs RA for each simulated MTJ. All configurations use Fe electrodes. The background gradient qualitatively illustrates MTJ performance. Red indicates regions of low TMR and high RA (poor performance), while green corresponds to high TMR and low RA (desirable performance). The black dashed line indicates the performance of the pure MgO MTJ as a benchmark for other structures, with the striped region highlighting those that outperform MgO.

to weaker evanescent attenuation and hence lower RA. For six-layer MgO (Figure 8a), the complex bands (Figure 8b) at Γ and X yield decay rates of Im(k) = 0.35 $\frac{2\pi}{a}$ and 0.63 $\frac{2\pi}{a}$, respectively. The relatively small Im(k) at Γ accounts for efficient tunneling of Δ_1-symmetric Fe majority states and the resulting high TMR in conventional MgO-based MTJs. The corresponding real band structure confirms the wide-gap insulating nature characteristic of MgO.

For 6-layer ScN (Figure 8c) the complex bandstructures (Figure 8d) at Γ and X reveal decay rates of Im(k) = 0.15 $\frac{2\pi}{a}$ and Im(k) = 0.44 $\frac{2\pi}{a}$, respectively. ScN's narrower gap and imaginary bands with flat dispersion in momentum space closer to Im(k) = 0, are consistent with its smaller decay constants. The lower decay rate of the six-layer ScN than that of the six-layer MgO structure reveals how ScN provides more electron transmission through the barrier for the same thickness than MgO. This explains the higher majority channel conduction in ScN and therefore lower RA. ScN and MgO are combined for the monolayer ScN and five-layer MgO configuration in Figure 8e. Its complex bandstructures (Figure 8f) show decay rates of Im(k) = 0.31 $\frac{2\pi}{a}$ at Γ and Im(k) = 0.62 $\frac{2\pi}{a}$ at X, slightly lower than those observed for the six-layer MgO structure. This suggests that for this hybrid ScN-MgO configuration, bulk tunneling attenuation is still governed by the five layers of MgO. This is further supported by the relatively low number of bands close to E_f compared to the six-layer ScN structure, while having a smaller bandgap than pristine MgO. The features here suggest that the single ScN layer introduces localized interface states without significantly altering the dispersive conduction pathways from MgO.

For six-layer TiN (Figure 8g), the complex bandstructures (Figure 8h) at Γ and X reveal decay rates of Im(k) = 0.12 $\frac{2\pi}{a}$ and Im(k) = 0.31 $\frac{2\pi}{a}$, respectively. Compared to MgO and ScN, the six-layer TiN barrier exhibits reduced decay constants at both Γ and X, and the complex-band arcs lie closer to Im(k) = 0. These features indicate weaker evanescent attenuation for a given thickness. Multiple low-attenuation pathways, as shown by bands crossing E_f and low Im(k) states, enhance transmission through TiN barriers. This combination explains the reduced TMR and lower RA of pristine TiN compared to its MgO and ScN counterparts. The 1TiN/5MgO configuration (Figure 8i) has decay rates of Im(k) = 0.26 $\frac{2\pi}{a}$ at Γ and Im(k) = 0.52 $\frac{2\pi}{a}$ at X (Figure 8j). The 1TiN/5MgO barrier exhibits decay constants intermediate between those of pure MgO and TiN and lower than the 1ScN/5MgO structure. With the dispersive bands originating from MgO and less dispersive bands from the monolayer, this feature suggests a hybridized state with contributions from both the MgO barrier and the TiN interface. A dispersive state intersects E_f in the real band section, consistent with the six-layer TiN structure. With respect to the six-layer MgO structure, The TiN insertion introduces additional states, modifying the band alignment relative to E_f, and lowering the decay constants. The lower decay rates suggest lower resistance in the hybrid TiN and ScN systems compared to MgO."

Figure 9 plots every simulated MTJ's TMR vs. RA. All structures are compared to the six-layer configuration of MgO (Fe/6MgO/Fe) which has a TMR ratio of 7,180%. Looking at the four pure-MgO benchmarks (4, 6, 8, 10 layers, indicated by the dashed line), their TMR rises steadily from 5,310% at four layers to 17,670% at ten layers, but the RA inflates by roughly two orders of magnitude. Replacing MgO with a full ScN barrier inverts this trade-off. Six, eight, and ten-layer ScN devices cut RA by one to two decades relative to equal-thickness MgO, yet their TMR values sit at 540%, 9,100%, and 760%, respectively. Notably, the eight-layer ScN sample has the lowest RA of any device above 5,000% TMR, with a ratio exceeding that of the six-layer MgO benchmark.

The pure six-layer NbN barrier has a low TMR of 30%, but 1NbN/4MgO/1NbN delivers 1,490% while keeping RA an order of magnitude below six-layer MgO. When increasing the oxide thickness to eight layers, the 1NbN/8NbN/1NbN structure shows a TMR of 8,490% with a greater RA than the ten-layer MgO structure. The 2NbN/2MgO/2NbN hybrids drop TMR to 390%. Relative to other results, ScN-MgO hybrids excel when the nitride is confined to a single atomic layer at each interface. The 1ScN/4MgO/1ScN junction achieves 14,260% TMR with an RA ten times lower than the six-layer MgO reference. Extending the MgO core to eight layers pushes TMR to 24,630% with only a modest RA increase. Doubling the dusting thickness (2ScN/2MgO/2ScN) or grouping nitride into two blocks (3MgO/3ScN) reduces TMR to 820% and 140%, respectively, confirming that thin nitride layers at the interface maximize spin selectivity. This is consistent with the structural variations seen in the hybrid TiN MTJs. TiN behaves like ScN at small thickness with 1TiN/4MgO/1TiN yielding 12,000% TMR. Beyond 10 layers, a twelve-layer stack (1TiN/10MgO/1TiN) reaches 32,310% TMR while exhibiting an RA that is nearly an order of magnitude higher than MgO of equivalent thickness with a similar TMR. Intriguingly, the 2TiN/4MgO/2TiN MTJ has a TMR ratio of 10,600% with an RA nearly an order of magnitude lower than the six-layer MgO benchmark, despite having 2 more total layers. The dusting layer VN structure achieves a similar TMR ratio (4,170%) to that of the MgO benchmark with an RA lower by 70%. 1TaN/4MgO/1TaN has a lower TMR than the dusting layer VN structure by 380% and a higher RA. Notably, six-layer ZnO has a TMR ratio of 6,600%, close to MgO benchmark, but an RA product nearly two orders of magnitude lower. SiN dusting (360%) and DyN dusting (140%) sit well inside the frontier. Their mid-range TMR and RA values mark them as second-tier candidates. Furthermore, barrier materials that fall far under the TMR performance benchmark include six-layer DyN (140%), 6-layer VN (100%), and six-layer TaN (80%). The performances of the various nitride dusting layer structures highlight the dominance of the ScN-MgO and TiN-MgO based devices as they consistently outperform the pristine MgO MTJs at various thicknesses, as illustrated by points in the striped shading region. The results map yields concrete design guidance, suggesting that monolayer nitride dusting of ScN or TiN is the most RA-efficient route to enhanced TMR. To verify that the observed improvements are specific to the LEAD-predicted materials and not a general consequence of nitride dusting, additional control hybrids were simulated under identical conditions and found to have low TMR (see Figure S3, Supporting Information).

The LEAD framework offers both scalability and computational efficiency advantages compared to conventional high throughput DFT workflows. Traditional high throughput studies, such as that by Kim et al.,[63] require hundreds of DFT evaluations, even after machine learning acceleration, to identify a few stable materials from millions of candidates. In contrast, LEAD's filter first approach uses literature trained language models to pre-rank candidates, reducing the total computational load by two to three orders of magnitude. The NLP stage required under 50 GPU hours, and DFT verification under 600 CPU core hours (computational costs of LEAD pipeline can be found in Section S3.4 of Supporting Information).

Beyond magnetic tunnel junctions, LEAD's foundation in text derived embedding relationships rather than task specific labels enables straightforward adaptation to other material and device classes. LEAD can be retrained or re-prompted on corpora targeting different topics such as ferroelectrics, superconductivity, inductor core materials, thermoelectric efficiency, or catalysis. Likewise, the same pipeline can prioritize interface combinations, heterostructure geometries, or dopant configurations.

4. Conclusion

This study demonstrates a data-driven route for discovering next-generation tunnel barriers by coupling LM inference with first-principles validation. Starting from a corpus of materials science abstracts, we trained Word2Vec and MaterialsBERT models that uncovered latent semantic links between MTJ relevant concepts and previously under explored compounds. Guided by these embeddings, we shortlisted nitrides, oxides, and hybrid structures for DFT screening in QuantumATK. The DFT transport calculations reveal that thin nitride dusting layers at the MgO interface can simultaneously lower the RA and boost TMR. A configuration featuring ScN monolayers at the interface attains a TMR of 14,260% (roughly 2 times the 6-layer MgO benchmark) with comparable RA, while Fe/1TiN/4MgO/1TiN/Fe delivers a TMR ≈12,000% with a slightly higher RA.

The success of the LEAD pipeline underscores the power of literature-trained language models to accelerate spintronic materials discovery. The results suggest nitride-modified barriers as a novel materials family for overcoming the scaling limits of MgO. Nevertheless, maintaining the dusting interface after annealing is a potential challenge. Future work should experimentally synthesize ScN and TiN dusting layer junctions to validate the predicted enhanced TMR and lower RA. An analysis of epitaxial strain and thermal stability for candidate barriers is provided in Table S7 (Supporting Information) to guide experimental feasibility.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

AI, magnetic tunnel junction, theory, tunnel magnetoresistance

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